



## Boring routine?

Is your daily routine boring? Watch this to see what you should not expect to come out.  
<http://dgit.in/In-Out>

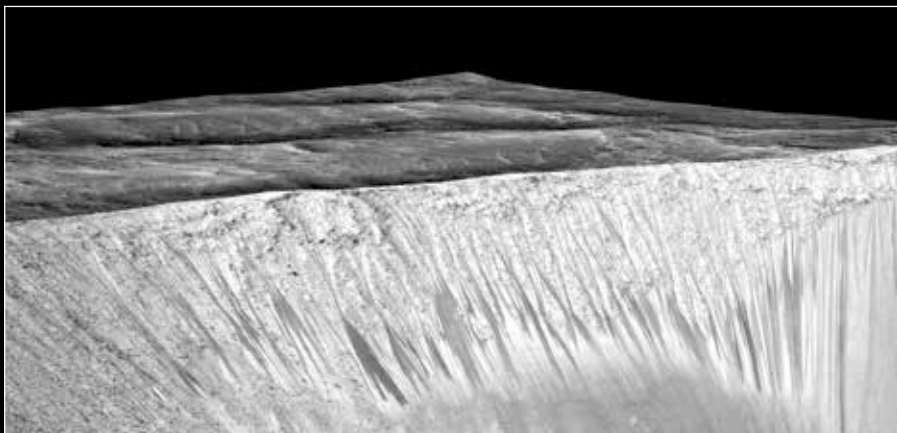


## Living dead: Plants!

Zombie plants also exist in nature. Find out how and why plants zombify.  
<http://dgit.in/ZombiePlant>

Space Age

IMAGE CREDIT: NASA/JPL-CALTECH/UNIV. OF ARIZONA



RSL / dark streaks on the walls of the Garni Crater on Mars. The streaks are a few hundred metres long, and have now proven to be caused by slow flowing water

than the 34% salinity of the Dead Sea. Normal ocean water has about 3.5% salinity levels; just in case you need a reference.

It's hard enough for life to survive with that amount of salinity, but add on the almost non-existent atmosphere of Mars (as compared to Earth's rather comfy atmosphere), and the fact that scientists expect the salinity of Mars seepage to be even higher than Don Juan, and you have a very weak case for finding life.

However, new studies seem to suggest that certain types of life forms can adapt to survive in water as salty as that which is found in the Don Juan Pond, which

makes the optimists let out a cheer! Either way, we're never going to find out until we go there – in person or in camera (through robots such as rovers). The challenge is not contaminating Mars (and these salty streaks) with life from Earth. Although a lot of care goes into trying to sterilize things for space, we all know that life is stubborn, and always finds a way...

If we ever send humans there, we have the additional headache of making sure that we don't bring any unknown life forms back to Earth with us. After all, just a single Martian virus could potentially wipe out all of Earth's life, or infect it.

## All your base...

One of the most important resources needed for space exploration also happens to be water. The fact that water is made up of Hydrogen and Oxygen, which is basically A-grade rocket fuel, is mostly the reason why it's so desired. Then again, the fact that we humans cannot live without very large amounts of it (not just to drink, but also to bathe, wash up, cleanse, dissolve stuff in, etc., also plays a large role in why we want water wherever we go. Another advantage of Mars is the perchlorates, because rockets currently use some perchlorates as rocket fuel.

All long range exploration needs food, water and fuel to be able to succeed. In terms of Mars, it's obviously desirable to get that water in an already liquid form, and not have to chip away chunks of ice from the frozen poles of Mars, and then melt and purify it. It's the reason why Europa, one of Jupiter's moons which is known to have a lot of water in the form of ice, would be a challenge if we tried to use it as a base station in the distant future. Mars certainly is the closest planet / moon with water on it, and that makes it the perfect candidate for the first off Earth base station in our attempts to explore the solar system and beyond.

## Just because!

Of course the most buzz around this discovery is to try and get people interested in getting off this rock and looking to the stars for a change, instead of the bleak over-crowded future we all face if we don't. If there's water there, maybe we can go there... that's the thought that many scientists want the general public to have. Because as we all know, unless the politicians – who control all the money – feel that the general public is interested in humans exploring planets other than our own, we're never going to get funding for such missions.

It's Mars. Another planet. It's far away. It might contain life. It might have fossils. It's the current frontier that we haven't been able to cross. Do we really need more reasons to want to go there? How about a super-chilled glass of grade-A brine? That settles it! Mars or bust!

IMAGE CREDIT: WIKIMEDIA



The Don Juan Pond in Antarctica is the world's saltiest water body, and even that may pale in comparison to the saltiness of the water on Mars!